Chapter 6: The 2^k Factorial Design

Factorial designs are widely used in experiments involving several factors where it is necessary to study the **joint effect** of the factors on response.

We discussed the general methods for the analysis of factorial designs in the last chapter. However, there are several special cases of general factorial design that are important, because they **widely** used in research work and also they form the **basis** of other designs of considerable practical value.

The $\mathbf{2^k}$ **factorial design** is the **most important** of these special cases.

This chapter focuses on this extremely important class of design.

Consider a factorial design with k factors, each at **only two** levels.

The levels may be **quantitative,** such as two values of temperature, pressure, or time; or they may be **qualitative**, such as two machines, two operators, the "high" and "low" level of factor, or perhaps the presence and absence of a factor.

A complete replicate of such a design requires $2 \times 2 \times \cdots \times 2 = 2^k$ observations and is called a $\mathbf{2^k}$ **factorial design**.

The design is particularly useful in the **early stage** of experimental work, when there are likely to be **many factors** to be investigated.

It provides the **smallest number of runs** with factors can be studied in a complete factorial design. Consequently, these designs are widely used in **factor screening experiments**.

The 2^2 Factorial Design: The first design in the 2^k series is one with only **two factors**, Say A and B, each run at **two levels**. This design is called a **2^2 factorial design**. The levels of the factors may be arbitrarily "low" and "high".

Example 1 (From textbook, page 234): A researcher investigated the effect of the concentration of the reactant and the amount of the catalyst on the conversion (yield) in a chemical process. Let the two levels of interest for reactant concentrations (factor A) were 15 percent and 25 percent. The catalyst was factor B, with high level denoting the use of two pounds of the catalyst and the low level denoting the use of only one pound. The experiment was replicated three times, and the data are given in the following table.

Factor A	Factor B	Treatment Combination	Replicate I	Replicate II	Replicate III	Total	Label
−	−	A low, B low	28	25	27	80	(1)
+	−	A high, B low	36	32	32	100	(a)
−	+	A low, B high	18	19	23	60	(b)
+	+	A high, B high	31	30	29	90	(ab)

Notations: It is often convenient to write down the treatment combinations in the order (1), a, b, and ab. This is referred to as **standard order**.

(1) **= sum of observations at low levels of both factors,**

a **= sum of observations with A high and B low,**

b **= sum of observations with B high and A low,**

ab **= sum of observations with both high.**

Statistical Model:

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk}, i, j = 1,2 \text{ and } k = 1,2, \ldots n$$

A_1= Main effect of low level of A and A_2= Main effect of high level of A

Since $A_1 + A_2 = 0$, we have $A_1 = -A_2$.

The **main effect** of A is define as: $A_2 - A_1$.

Example 1:

Factor A	Factor B	Treatment Combination	Replicate I	Replicate II	Replicate III	Total	Label
–	–	A low, B low	28	25	27	80	(1)
+	–	A high, B low	36	32	32	100	(a)
–	+	A low, B high	18	19	23	60	(b)
+	+	A high, B high	31	30	29	90	(ab)

$$A = \frac{1}{2n}\left[(ab-b)+(a-1)\right] = \frac{1}{2n}\left[ab+a-b-(1)\right] = \frac{ab+a}{2n} - \frac{b+1}{2n} = \bar{y}_{A^+} - \bar{y}_{A^-}$$

Main effect of A: Average of y at high A – Average of y at low A

$$A = \frac{1}{2(3)}(90+100-60-80) = 8.33$$

The **effect of A** is **positive**; this suggests that **increasing** A from low level (15%) to the high level (25%) will **increase** the yield.

Example 1:

Factor A	Factor B	Treatment Combination	Replicate I	Replicate II	Replicate III	Total	Label
–	–	A low, B low	28	25	27	80	(1)
+	–	A high, B low	36	32	32	100	(a)
–	+	A low, B high	18	19	23	60	(b)
+	+	A high, B high	31	30	29	90	(ab)

$$B = \frac{1}{2n}\left[(ab-a)+(b-1)\right] = \frac{1}{2n}\left[ab+b-a-(1)\right] = \frac{ab+b}{2n} - \frac{a+1}{2n} = \bar{y}_{B^+} - \bar{y}_{B^-}$$

Main effect of B: Average of y at high B – Average of y at low B

$$B = \frac{1}{2(3)}(90+60-100-80) = -5.00$$

The **effect of B** is **negative**; this suggests that **increasing** the amount of catalyst added to the process will **decrease** the yield.

Example 1:

Factor A	Factor B	Treatment Combination	Replicate I	Replicate II	Replicate III	Total	Label
−	−	A low, B low	28	25	27	80	(1)
+	−	A high, B low	36	32	32	100	(a)
−	+	A low, B high	18	19	23	60	(b)
+	+	A high, B high	31	30	29	90	(ab)

The **interaction effect AB** is the average **differences** between the effect of A at the high level of B and the effect of A at the low level of B.

$$AB = \frac{1}{2n}\left[(ab-b)-(a-1)\right] = \frac{1}{2n}\left[ab+1-a-b\right] = \frac{ab+1}{2n} - \frac{a+b}{2n}$$

Alternatively, the **interaction effect** is the average differences between the effect of B at the high level of A and the effect of B at the low level of A

$$AB = \frac{1}{2n}\left[(ab-a)-(b-1)\right] = \frac{1}{2n}\left[ab+1-a-b\right] = \frac{ab+1}{2n} - \frac{a+b}{2n}$$

$$AB = \frac{1}{2(3)}(90+80-100-60) = 1.67$$

The interaction effect appears to be small relative to the two main effects.

Contrasts:

$$A = \frac{1}{2n}\left[(ab-b)+(a-1)\right] = \frac{ab+a-b-1}{2n} = \frac{C_A}{2n}$$

$$B = \frac{1}{2n}\left[(ab-a)+(b-1)\right] = \frac{ab+b-a-1}{2n} = \frac{C_B}{2n}$$

$$AB = \frac{1}{2n}\left[(ab-b)-(a-1)\right] = \frac{ab+1-a-b}{2n} = \frac{C_{AB}}{2n}$$

The contrast C_A, C_B and C_{AB} are **orthogonal** contrasts. It can be shown that:

$$SS_A = \frac{C_A^2}{4n},\ SS_B = \frac{C_B^2}{4n}$$

and

$$SS_{AB} = \frac{C_{AB}^2}{4n}$$

Example 1:

Factor A	Factor B	Treatment Combination	Replicate I	Replicate II	Replicate III	Total	Label
−	−	A low, B low	28	25	27	80	(1)
+	−	A high, B low	36	32	32	100	(a)
−	+	A low, B high	18	19	23	60	(b)
+	+	A high, B high	31	30	29	90	(ab)

$$SS_A = \frac{C_A^2}{4n}, SS_B = \frac{C_B^2}{4n} \text{ and } SS_{AB} = \frac{C_{AB}^2}{4n}$$

$$SS_A = \frac{C_A^2}{4n} = \frac{(90 + 100 - 60 - 80)^2}{4(3)} = 208.3$$

$$SS_B = \frac{C_B^2}{4n} = \frac{(90 + 60 - 100 - 80)^2}{4(3)} = 75$$

$$SS_{AB} = \frac{C_{AB}^2}{4n} = \frac{(90 + 80 - 100 - 60)^2}{4(3)} = 8.33$$

Since $SS_{Total} = 323.00$, we have

$$SS_{Error} = SS_{Total} - SS_A - SS_B - SS_{AB}$$
$$= 323 - 208.33 - 75 - 8.3331.34$$

The **ANOVA table**:

Source	df	SS	MS	F
A	1	208.33	208.33	53.2
B	1	75.00	75.00	19.1
AB	1	8.33	8.33	2.1
Error	8	31.44	3.92	
Total	11	323.00		

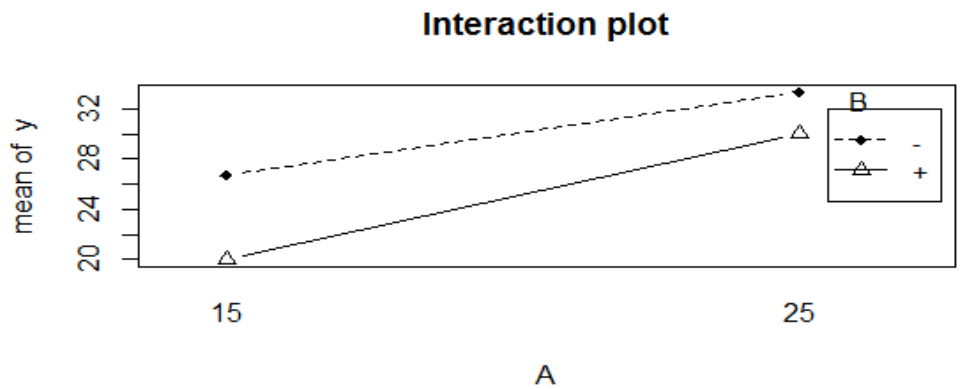
R Codes to Find Interaction (or Profile) plots:

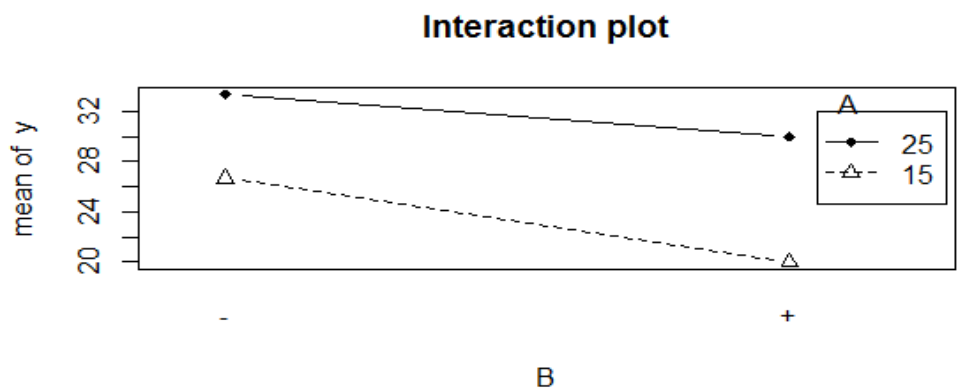
```
> y= c(28, 36, 18, 31, 25, 32, 19, 30, 27, 32, 23, 29)
> A = factor(rep(c(15,25,15,25), times=3))
> B = factor(rep(c("-", "+"), each= 2, times=3))

> par(mfrow=c(2,1))

> interaction.plot(A,B,y,
   type="b", pch=c(18,24),
   leg.bty="o",
   main = "Interaction plot" )

> interaction.plot(B,A,y,
    type="b", pch=c(24,18),
     leg.bty="o",
     main = "Interaction plot" )
```





R Codes to Obtain ANOVA Table for the Factorial Design:

```
> model <- lm(y ~ A +B + A*B)
> anova(model)
```

Analysis of Variance Table

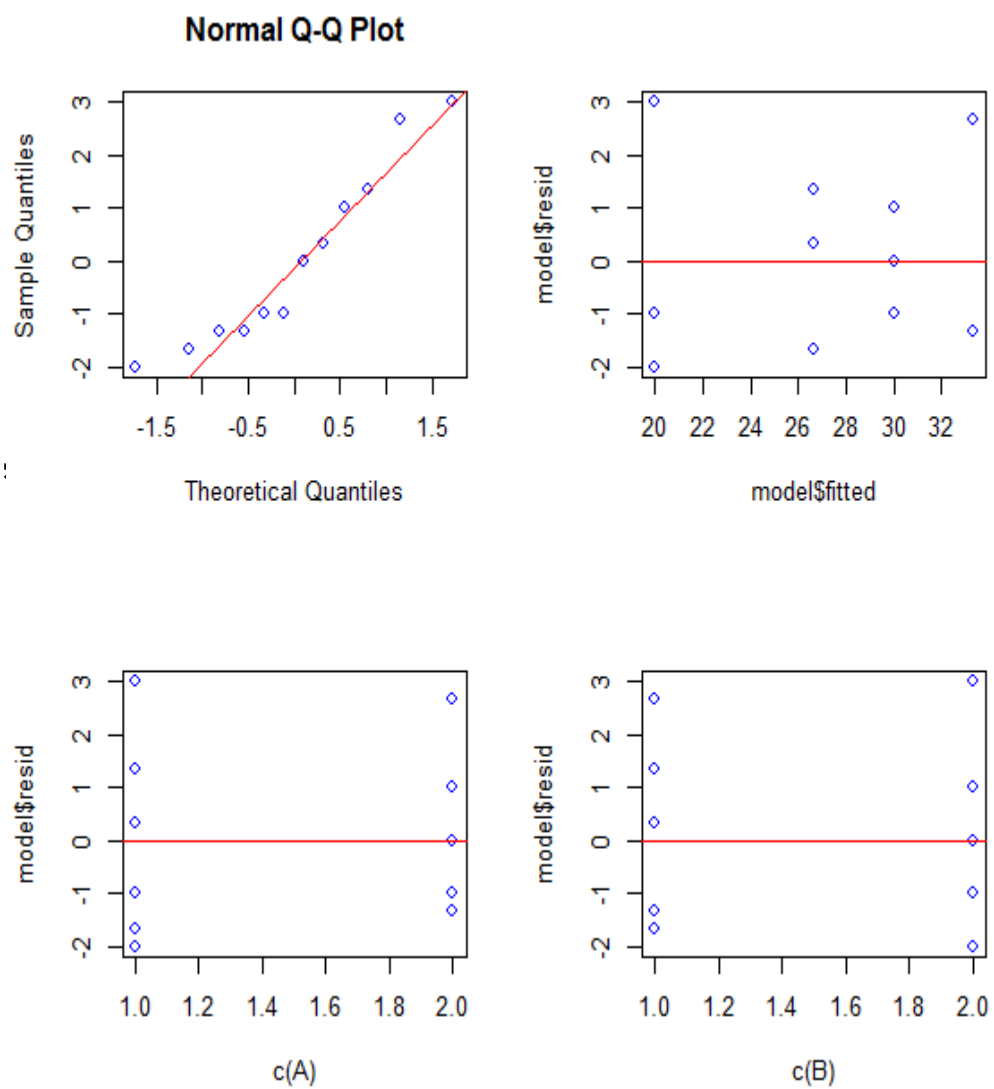
Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	1	208.333	208.333	53.1915	8.444e-05 ***
B	1	75.000	75.000	19.1489	0.002362 **
A:B	1	8.333	8.333	2.1277	0.182776
Residuals	8	31.333	3.917		

Signif. codes: 0 ‘*’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1**

R Codes to Check Assumptions:

```
> windows()
> par(mfrow=c(2,2))
> qqnorm(model$resid,
    col="blue")
> qqline(model$resid, col=2)
> plot(model$fitted,model$resid,
    col="blue")
> abline(h=0, col="red")
> plot(c(type), model$resid,
        col="blue")
> abline(h=0, col="red")
>plot(c(temp), model$resid,
        col="blue")
>abline(h=0, col="red")
```



Regression Equation: Suppose variable x_1 is the **coded variable** that represents the reactant concentration. Therefore, If the concentration is at the high level (25%), then $x_1 = 1$; If the concentration is at the low level (15%), then $x_1 = -1$.

Similarly, suppose x_2 is a **coded variable** that represent the amount of catalyst. Therefore, If the catalyst is at the high level (2 pounds), then $x_2 = 1$; If the catalyst is at the low level (1 pound), then $x_2 = -1$.

The fitted regression model is:

$$\begin{aligned}\hat{y} &= \bar{y}_{\ldots} + \frac{Effect\ of\ A}{2}x_1 + \frac{Effect\ of\ B}{2}x_2 \\ &= 27.5 + \frac{8.33}{2}x_1 + \frac{-5}{2}x_2 \\ &= 2.75 + 4.167x_1 - 2.5x_2\end{aligned}$$

R Codes to Obtain Regression Equation:

```
> y= c(28, 36, 18, 31, 25, 32, 19, 30, 27, 32, 23, 29)
> x1 = rep(c(-1,1,-1,1), times=3)
> x2 = rep(c(-1,1), each= 2, times=3)
> model1= lm(y~x1+x2)
> model1
```

Call:

lm(formula = y ~ x1 + x2)

Coefficients:

(Intercept)	x1	x2
27.500	4.167	-2.500

Regression Equation: Suppose variable x_1 is the **coded variable** that represents the reactant concentration. Therefore, If the concentration is at the high level (25%), then $x_1 = 1$; If the concentration is at the low level (15%), then $x_1 = -1$.

$$x_1 = \frac{\text{Concentration} - (25+15)/2}{(25-15)/2} = \frac{\text{Concentration} - 20}{5}$$

Similarly, suppose x_2 is a **coded variable** that represent the amount of catalyst. Therefore, If the catalyst is at the high level (2 pounds), then $x_2 = 1$; If the catalyst is at the low level (1 pound), then $x_2 = -1$.

$$x_2 = \frac{\text{Catalyst} - (2+1)/2}{(2-1)/2} = \frac{\text{Catalyst} - 1.5}{0.5}$$

The fitted regression model is:

$$\hat{y} == 2.75 + 4.167\left(\frac{\text{Concentration}-20}{5}\right) - 2.5\left(\frac{\text{Catalyst}-1.5}{0.5}\right)$$
$$= 18.33 + 0.8333\ \text{Concentration} - 5.00\ \text{Catalyst}$$

R Codes to Obtain Regression Equation:

```
> y= c(28, 36, 18, 31, 25, 32, 19, 30, 27, 32, 23, 29)
> concentration = rep(c(15,25,15,25), times=3)
> catalyst = rep(c(1,2), each= 2, times=3)
> model2= lm(y~concentration+catalyst)
> model2
```

Call:

lm(formula = y ~ concentration + catalyst)

Coefficients:

(Intercept)	concentration	catalyst
18.3333	0.8333	-5.0000

Using the standard order, we see that the contrast coefficient used in estimating the effect are:

Effects	(1)	a	b	ab
A	-1	+1	-1	+1
B	-1	-1	+1	+1
AB	+1	-1	-1	+1

Note: The **contrast coefficients** for estimating the interaction effect are just the product of the product of the corresponding coefficient for the two main effects. The contrast coefficient is always either +1 or −1, and a table of plus and minus sign such as the following table can be used to determine the proper sign for each treatment combination.

Treatment Combination	Factorial Effect			
	I	A	B	AB
(1)	+	−	−	+
a	+	+	−	−
b	+	−	+	−
ab	+	+	+	+

The 2^3 Design: Suppose that there are **three** factors, A, B, and C, each at **two levels**, are of interest. The design is called a **2^3 Factorial design**.

There are **three** different notations that are widely used for the runs in the design.

The first is $+$ and $-$ notation, often called **geometric notation**. The eight runs in the design are listed in the a table and this is sometimes called the **design matrix**.

The second is the use of lowercase letter labels in **standard order** as (1), a, b, ab, c, ac, bc, and abc for the design.

The final notation uses 1 and 0 to denote high and low factor level, respectively, instead of $+$ and $-$ notation.

These three different notations are illustrated in the following table for the design.

Run	A	B	C	Labels	A	B	C
1	–	–	–	(1)	0	0	0
2	+	–	–	a	1	0	0
3	–	+	–	b	0	1	0
4	+	+	–	ab	1	1	0
5	–	–	+	c	0	0	1
6	+	–	+	ac	1	0	1
7	–	+	+	bc	0	1	1
8	+	+	+	abc	1	1	1

The Main Effect of A:

The **effect of A** when both **B** and **C** are at the **low** level is $\frac{a-(1)}{n}$.

The **effect of A** when **B** is at **high** level and **C** is at the **low** level is $\frac{ab-b}{n}$.

The **effect of A** when **B** is at the **low** level and **C** is at the **high** level is $\frac{ac-c}{n}$.

The **effect of A** when both **B** and **C** are at the **high** is $\frac{abc-bc}{n}$.

The **effect of A** is just the average of the above four effects.

$$A = \frac{1}{4n}\left[(a-(1)) + (ab-b) + (ac-c) + (abc-bc)\right]$$

$$= \frac{a+ab+ac+abc}{4n} - \frac{(1)+b+c+bc}{4n} = \bar{y}_{A^+} - \bar{y}_{A^-}$$

The Main Effect of B:

The **effect of B** when both **A** and **C** are at the **low** level is $\frac{b-(1)}{n}$.

The **effect of B** when **A** is at **high** level and **C** is at the **low** level is $\frac{ab-a}{n}$.

The **effect of B** when **A** is at the **low** level and **C** is at the **high** level is $\frac{bc-c}{n}$.

The **effect of B** when both **A** and **C** are at the **high** is $\frac{abc-ac}{n}$.

The **effect of B** is just the average of the above four effects.

$$B = \frac{1}{4n}\left[(b-(1)) + (ab-a) + (bc-c) + (abc-ac)\right]$$

$$= \frac{b+ab+bc+abc}{4n} - \frac{(1)+a+c+ac}{4n} = \bar{y}_{B^+} - \bar{y}_{B^-}$$

The Main Effect of C:

The **effect of C** when both **A** and **B** are at the **low** level is $\frac{c-(1)}{n}$.

The **effect of C** when **A** is at **high** level and **B** is at the **low** level is $\frac{ac-a}{n}$.

The **effect of C** when **A** is at the **low** level and **B** is at the **high** level is $\frac{bc-b}{n}$.

The **effect of C** when both **A** and **B** are at the **high** is $\frac{abc-ab}{n}$.

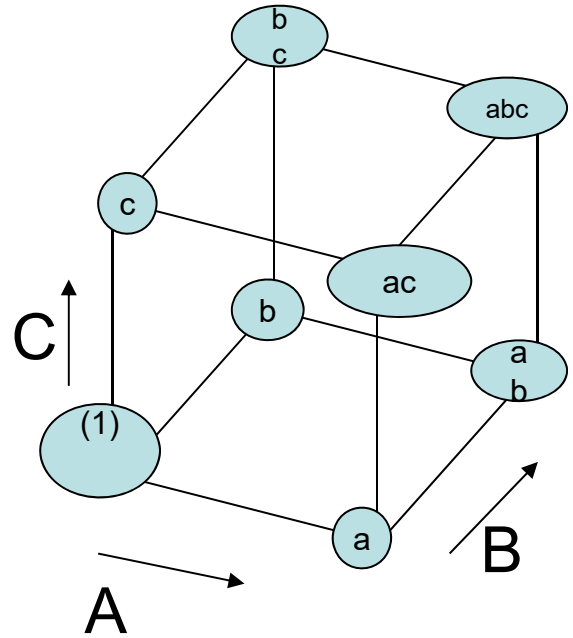
The **effect of C** is just the average of the above four effects.

$$C = \frac{1}{4n}\left[(c-(1)) + (ac-a) + (bc-b) + (abc-ab)\right]$$

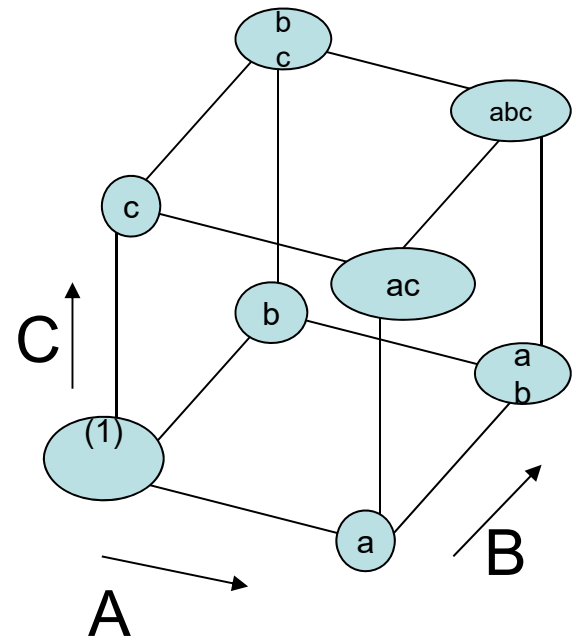
$$= \frac{c+ac+bc+abc}{4n} - \frac{(1)+a+b+ab}{4n} = \bar{y}_{C^+} - \bar{y}_{C^-}$$

The Main Effects:

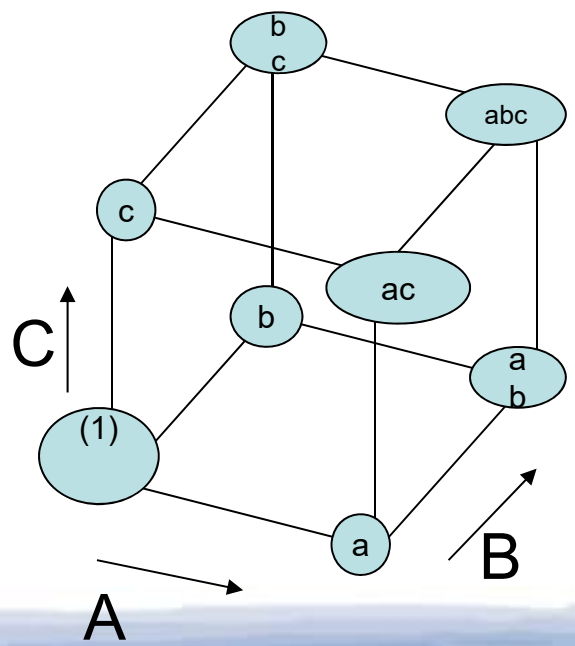
$$A = \bar{y}_{A^+} - \bar{y}_{A^-},$$



$$B = \bar{y}_{B^+} - \bar{y}_{B^-},$$



$$C = \bar{y}_{C^+} - \bar{y}_{C^-}$$



The Effect of AB:

The average of **A** effect when **B** is at the **high** level is .

$$\frac{(abc-bc)+(ab-b)}{2n}$$

The average of **A** effect when **B** is at the **low** level is .

$$\frac{(ac-c)+(a-(1))}{2n}$$

The **AB interaction** is just one-half of the difference of the above Effects.

$$\mathrm{AB}=\frac{1}{2}\left\{\frac{[(abc-bc)+(ab-b)]}{2n}-\frac{[(ac-c)+(a-(1))]}{2n}\right\}$$

$$=\frac{(abc-bc)+(ab-b)-(ac-c)-(a-(1))}{4n}$$

$$=\frac{abc+ab+c+(1)}{4n}-\frac{bc+b+ac+a}{4n}$$

The Effect of AC:

The average of **A** effect when **C** is at the **high** level is .

$$\frac{(abc-bc)+(ac-c)}{2n}$$

The average of **A** effect when **C** is at the **low** level is .

$$\frac{(ab-b)+(a-(1))}{2n}$$

The **AC interaction** is just one-half of the difference of the above Effects.

$$\mathrm{AC}=\frac{1}{2}\left\{\frac{[(abc-bc)+(ac-c)]}{2n}-\frac{[(ab-b)+(a-(1))]}{2n}\right\}$$

$$=\frac{(abc-bc)+(ac-c)-(ab-b)-(a-(1))}{4n}$$

$$=\frac{abc+ac+b+(1)}{4n}-\frac{bc+c+ab+a}{4n}$$

The Effect of BC:

The average of **B** effect when **C** is at the **high** level is .

$$\frac{(abc-ac)+(bc-c)}{2n}$$

The average of **B** effect when **C** is at the **low** level is .

$$\frac{(ab-a)+(b-(1))}{2n}$$

The **BC interaction** is just one-half of the difference of the above Effects.

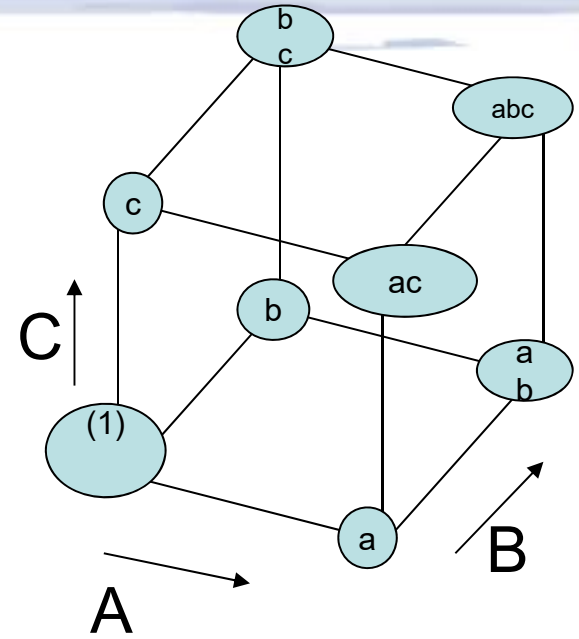
$$\mathrm{BC}=\frac{1}{2}\left\{\frac{[(abc-ac)+(bc-c)]}{2n}-\frac{[(ab-a)+(b-(1))]}{2n}\right\}$$

$$=\frac{(abc-ac)+(bc-c)-(ab-a)-(b-(1))}{4n}$$

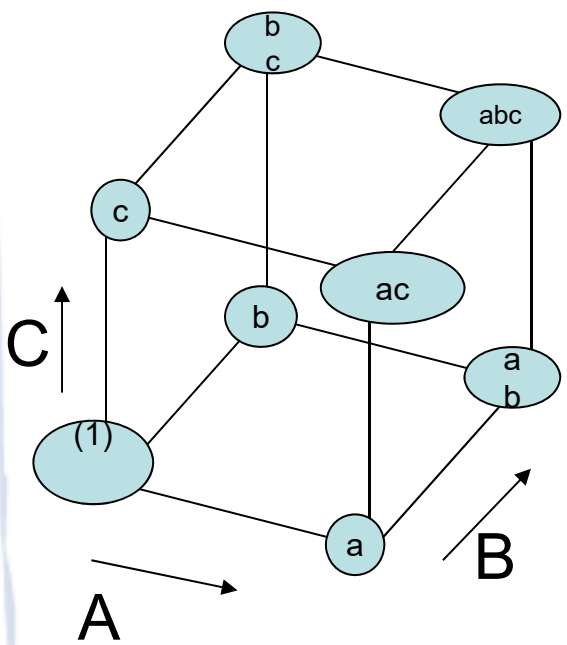
$$=\frac{abc+bc+a+(1)}{4n}-\frac{ac+c+ab+b}{4n}$$

The Interaction Effects:

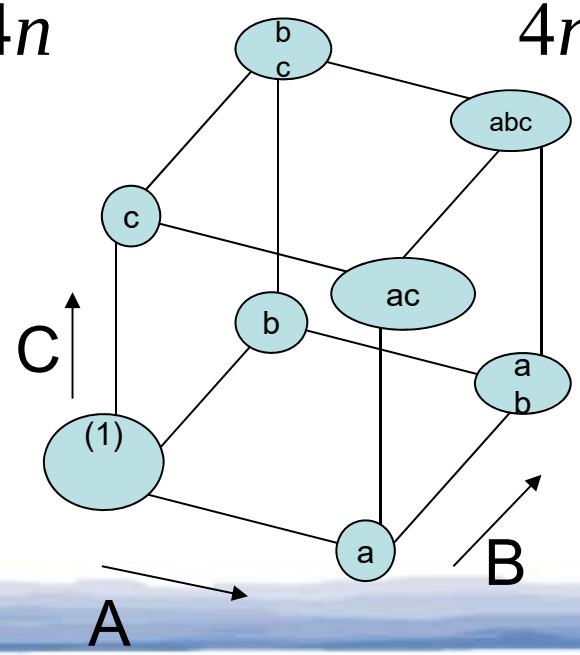
$$AB = \frac{abc + ab + c + (1)}{4n} - \frac{bc + b + ac + a}{4n}$$



$$AC = \frac{abc + ac + b + (1)}{4n} - \frac{bc + c + ab + a}{4n}$$



$$BC = \frac{abc + bc + a + (1)}{4n} - \frac{ac + c + ab + b}{4n}$$



The Effect of ABC:

The **AB interaction** effect when **C** is at the **high** level is .

$$\frac{(abc-bc)-(ac-c)}{2n}$$

The **AB interaction** effect when **C** is at the **low** level is .

$$\frac{(ab-b)-(a-(1))}{2n}$$

The **ABC interaction** is defined as the average difference between the AB interactions for the two difference levels of C. Thus

$$ABC = \frac{1}{4n}\left[(abc-bc)-(ac-c)-(ab-b)+(a-(1))\right]$$

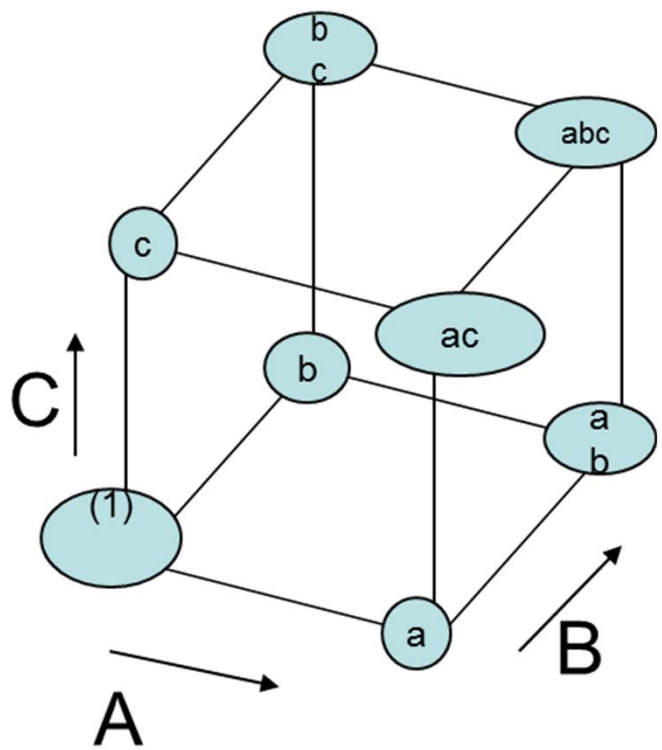
$$= \frac{1}{4n}\left[abc-bc-ac+c-ab+b+a-(1)\right]$$

$$= \frac{abc+c+b+a}{4n} - \frac{bc+ac+ab+(1)}{4n}$$

The Effect of ABC:

$$ABC = \frac{abc + c + b + a}{4n} - \frac{bc + ac + ab + (1)}{4n}$$

If the runs in the two averages are isolated, they define the vertices of the two **tetrahedral** that comprise the cube in the following picture.



To easily write down the estimates of he effects and the contrast, we start with a table of $\pm$ sign, done here for $k = 3$.

Treatment Combinations	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

The table of $\pm$ sign has several interesting properties:

a) Except for column ***I***, every column has an **equal** number of **plus** and **minus sign**.

b) The **sum** of the products of the signs in any two columns is **zero**.

c) Column I times any column leaves that column unchanged. That is, ***I*** is an **identity element**.

d) The **product** of any **two column** yields a column in the table. For example

$$A \times B = AB \text{ and } AB \times B = AB^2 = A$$

Example 2: The effect of percent carbonation, operating pressure, and line speed on the fill height of a carbonated beverage was studied in example 6 of chapter 5 which was example 5.3 from the textbook. Suppose that only two levels of carbonation are used so that the experiment is a factorial design with two replicates. The data are given in the following table.

	Operating Pressure (B)			
	25 psi		30 psi	
Percent Carbonation (A)	Line Speed (C) 200	250	Line Speed (C) 200	250
10	-3	-1	-1	1
	-1	0	0	1
	-4 = (1)	-1 = c	-1 =b	2 = bc
12	0	2	2	6
	1	1	3	5
	1 = a	3 = ac	5 = ab	11 = abc

Example 2:

$$A = \frac{1 + 5 + 3 + 11 - (-4) - (-1) - (-1) - 2}{8} = 3.00$$

$$B = \frac{1 + 5 + 2 + 11 - (-4) - 1 - (-1) - 3}{8} = 2.25$$

$$C = \frac{-1 + 3 + 2 + 11 - (-4) - 1 - (-1) - 5}{8} = 1.75$$

$$AB = \frac{5 - 1 - (-1) + (-4) + 11 - 2 - 3 + (-1)}{8} = 0.75$$

$$AC = \frac{-4 - 1 + (-1) - 5 - (-1) + 3 - 2 + 11}{8} = 0.25$$

$$BC = \frac{(-4) + 1 - (-1) - 5 - (-1) - 3 + 2 + 11}{8} = 0.50$$

$$ABC = \frac{11 - 2 - 3 + (-1) - 5 + (-1) + 1 - (-4)}{8} = 0.50$$

Sum of squares for the effects are easily computed, since each effect has a corresponding single degrees of freedom contrast. In a 2^kdesign with n replicates, the sum of squares for any effect is:

$$SS = \frac{(Contrast)^2}{2^k n}$$

Yates' Algorithm for Design:

Treatment Combinations	Response	(1)	(2)	(3)	Effect	Estimate of effect	Sum of Squares
(1)	-4	-3	1	16	I	--------	--------
a	1	4	15	24	A	3.00	36.00
b	-1	2	11	18	B	2.25	20.25
ab	5	13	13	6	AB	0.75	2.25
c	-1	5	7	14	C	1.75	12.25
ac	3	6	11	2	AC	0.25	0.25
bc	2	4	1	4	BC	0.50	1.00
abc	11	9	5	4	ABC	0.50	1.00

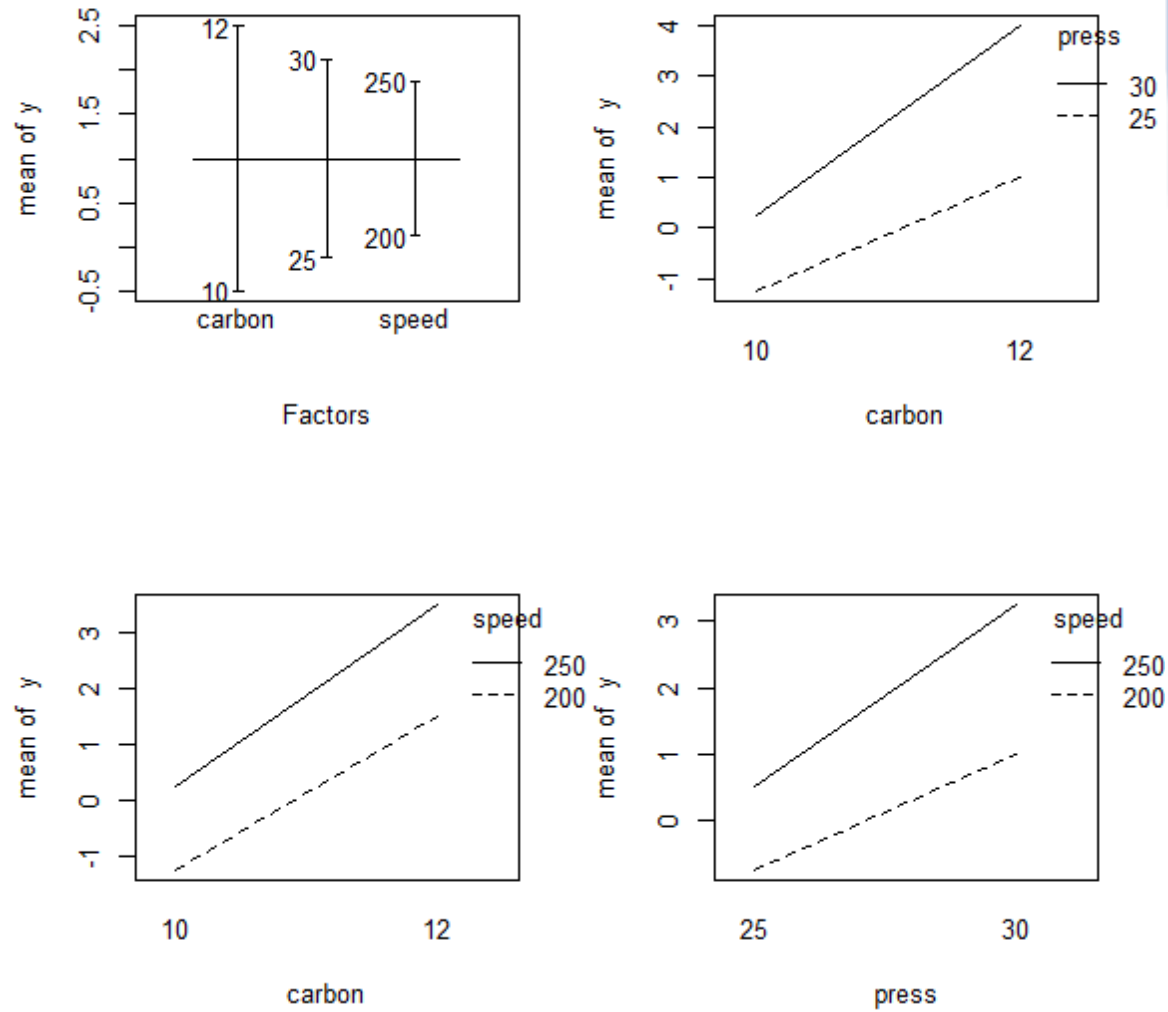
Example 2 (Cont.):

Percent Carbonation (A)	Operating Pressure (B) 25 psi — Line Speed (C) 200	25 psi — 250	30 psi — Line Speed (C) 200	30 psi — 250
10	-3	-1	-1	1
	-1	0	0	1
	-4 = (1)	-1 = c	-1 =b	2 = bc
12	0	2	2	6
	1	1	3	5
	1 = a	3 = ac	5 = ab	11 = abc

```
> y <- c(-3,-1,0,1, -1,0,2,1, -1,0,2,3, 1,1,6,5)
> carbon <- as.factor(rep(c(10,12),each=2, times=4))
> press <- as.factor(rep(c(25,30), each=8))
> speed <- as.factor(rep(c(200,250), each=4, times=2))
> data <- data.frame(y, carbon, press, speed)
> attach(data)
```

R Codes to Find Interaction Plot:

```
> par(mfrow=c(2,2))
> plot.design(data)
> interaction.plot(carbon,
                   press,y)
> interaction.plot(carbon,
                   speed,y)
> interaction.plot(press,
                   speed,y)
```



R Codes to Find ANOVA for CRFD:

```
> model <- lm(y ~ (carbon + press + speed )^3)
> anova(model)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
carbon	1	36.00	36.000	57.6	6.368e-05	***
press	1	20.25	20.250	32.4	0.0004585	***
speed	1	12.25	12.250	19.6	0.0022053	**
carbon:press	1	2.25	2.250	3.6	0.0943498	.
carbon:speed	1	0.25	0.250	0.4	0.5447373	
press:speed	1	1.00	1.000	1.6	0.2415040	
carbon:press:speed	1	1.00	1.000	1.6	0.2415040	
Residuals	8	5.00	0.625			

R Codes to Obtain Effects: To obtain the estimates of effects in R, you can use **model$effects** or **effects(model)**.

```
> model = lm(y~(A+B+C)^3)

> model.effects = model$effects[2:8]

>model.effects
```

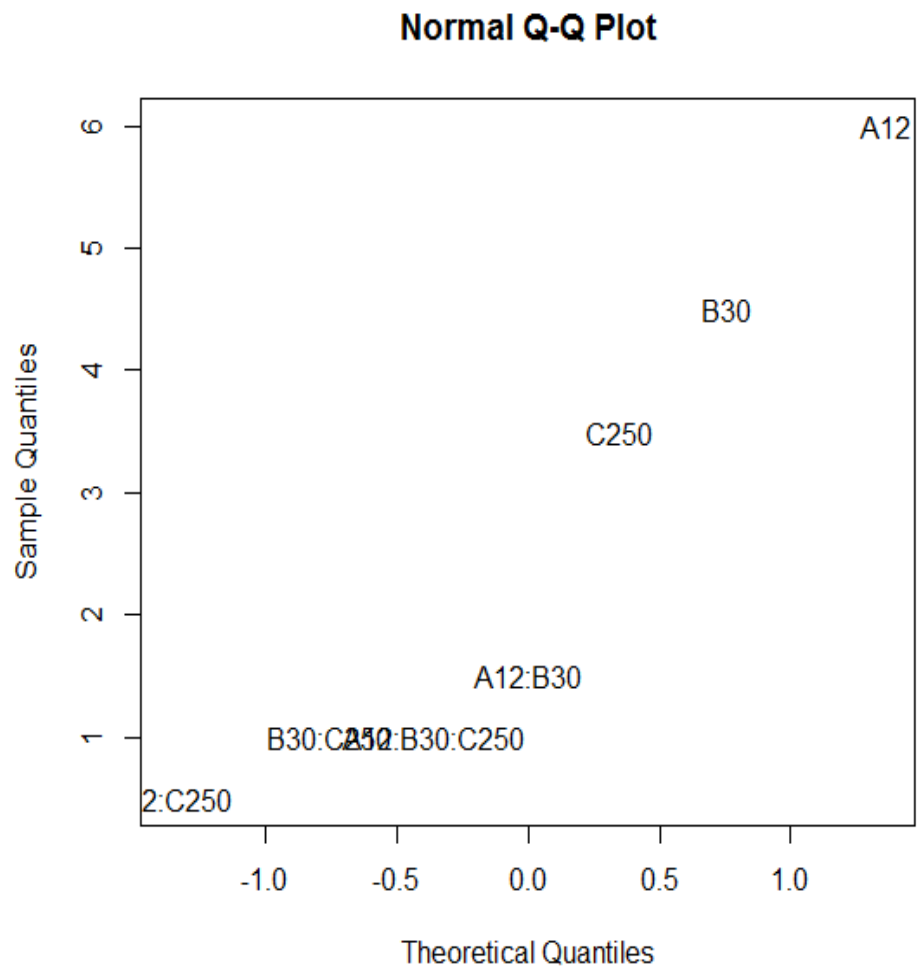
A12	B30	C250	A12:B30	A12:C250
6.0	4.5	3.5	-1.5	0.5

B30:C250	A12:B30:C250
1.0	1.0

Note: The codes gives you estimates that are twice as large in absolute value as those we obtained by hand calculations, and the signs sometimes differ. This is because of R definitions of "effect" and make no difference for comparing their **absolute** values.

Half-Normal Plot: This is plot of **absolute value** of the **effect** estimates against their cumulative normal probabilities.

```
> model.effects =
   model$effects[2:8]
> abs.effects =
    abs(model.effects)
> qq= qqnorm(abs.effects,
      type="n")
> text(qq$x,qq$y,
    labels=names(abs.effects))
```



The significant terms seems to be A, B, and C.

R Codes to Find ANOVA for CRFD:

```
> model1 <- lm(y ~ carbon + press + speed )
> anova(model1)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
carbon	1	36.00	36.000	45.474	2.063e-05 ***
press	1	20.25	20.250	25.579	0.000281 ***
speed	1	12.25	12.250	15.474	0.001986 **
Residuals	12	9.50	0.792		

R Codes to Compare Two Models:

```
> model <- lm(y ~ (carbon + press + speed )^3)
> model1 = lm(y~carbon+press+speed)
> anova(model1, model)
```

Analysis of Variance Table

Model 1: y ~ carbon + press + speed
Model 2: y ~ (carbon + press + speed)^3

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	12	9.5				
2	8	5.0	4	4.5	1.8	0.2221

The Regression Model:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 = \hat{\beta}_0 + Ax_1 + Bx_2 + Cx_3$$
$$= -2.5 + 3x_1 + 2.25x_2 + 1.75x_3$$

$$S.E.(\hat{\beta}_i) = S.E.(\text{Effect}) = S_{\text{Pooled}}\sqrt{\frac{1}{n2^{k-2}}},$$

$$\text{where } S_{\text{Pooled}} = \sqrt{MS(\text{Residual})}$$

$$S.E.(\hat{\beta}_i) = S_{\text{Pooled}}\sqrt{\frac{1}{n2^{k-2}}} = \sqrt{0.792}\sqrt{\frac{1}{2\times 2^{3-2}}} = 0.4449$$

The Regression Model:

> summary(model1)

Call:

lm(formula = y ~ carbon + press + speed)

$$\hat{y} = -2.5 + 3.00x_1 + 2.25x_2 + 1.75x_3$$

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-2.5000	0.4449	-5.620	0.000113 ***
carbon12	3.0000	0.4449	6.743	2.06e-05 ***
press30	2.2500	0.4449	5.058	0.000281 ***
speed250	1.7500	0.4449	3.934	0.001986 **

Un-Replicated Factorial Design: Example 6.2 from textbook

Run Number	Run Label	Filtration Rate (gal/h)
1	(1)	45
2	a	71
3	b	48
4	ab	65
5	c	68
6	ac	60
7	bc	80
8	abc	65
9	d	43
10	ad	100
11	bd	45
12	abd	104
13	cd	75
14	acd	86
15	bcd	70
16	abcd	96

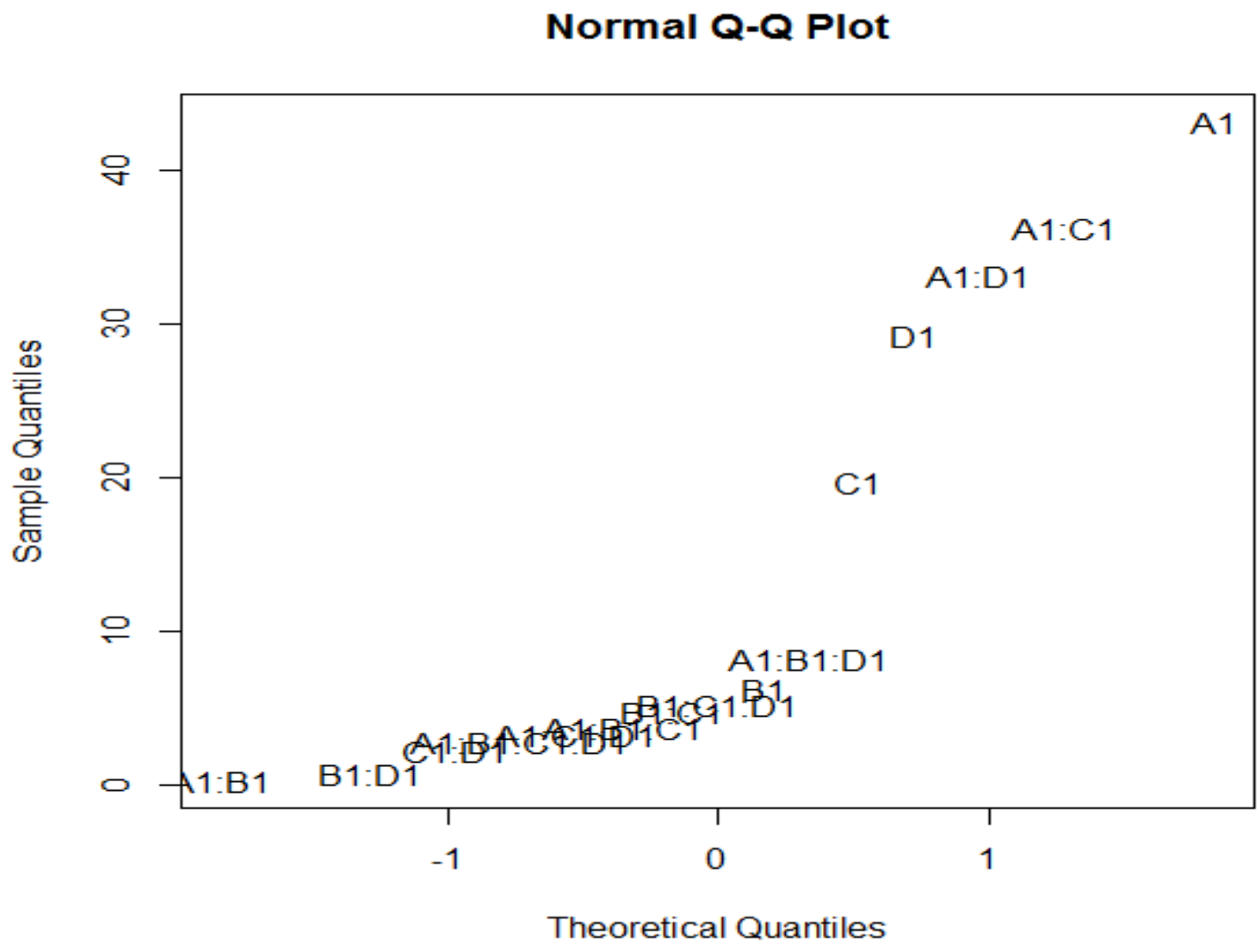
All the R codes can be found on Eclass which is obtained from Dr. Wiens's website

```
>  g <- lm(y ~ (A+B+C+D)^4)        > anova(g)
```

Analysis of Variance Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	**1**	**1870.56**	**1870.56**		
B	**1**	**39.06**	**39.06**		
C	**1**	**390.06**	**390.06**		
D	**1**	**855.56**	**855.56**		
.	**.**	**…..**	**……**		
.	**.**	**…...**	**……..**		
Residuals	**0**	**0.00**			

Warning message: In anova.lm(g) : ANOVA F-tests on an essentially perfect fit are unreliable


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
0
10
20
30
40
-1
0
1
A1
A1:C1
A1:D1
D1
C1
A1:B1:D1
B1
B1:C1:D1
A1:B1
B1:D1

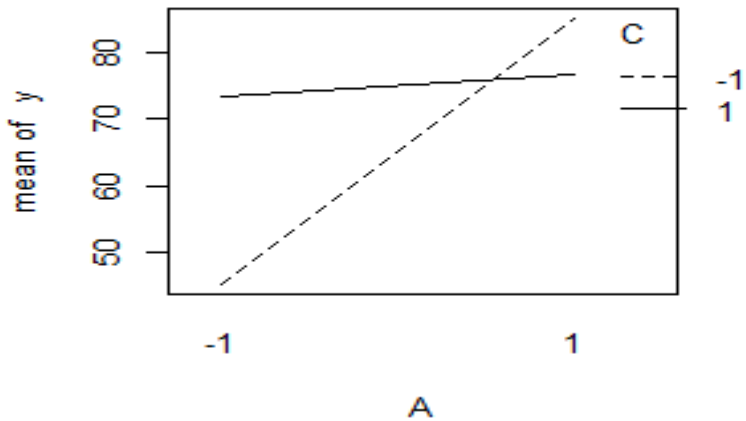
Let us drop B and fit the model with all terms not involving B.

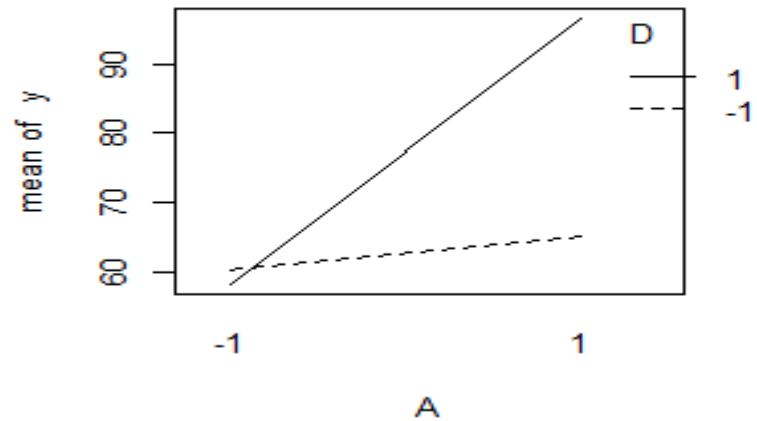
```
> h <- lm(y ~ (A+C+D)^3)  > anova(h)
```

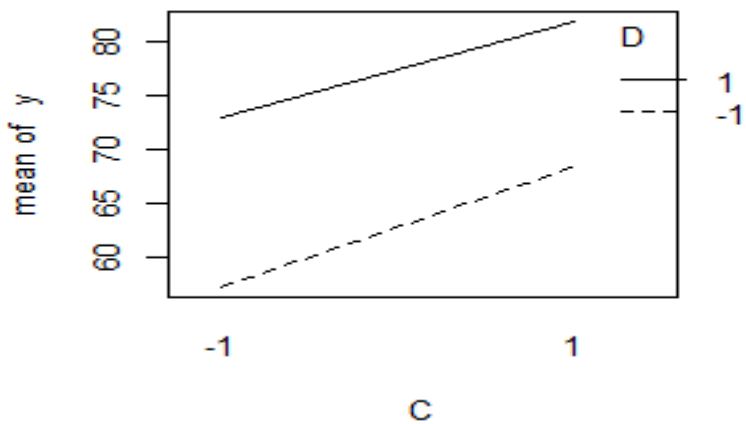
Analysis of Variance Table

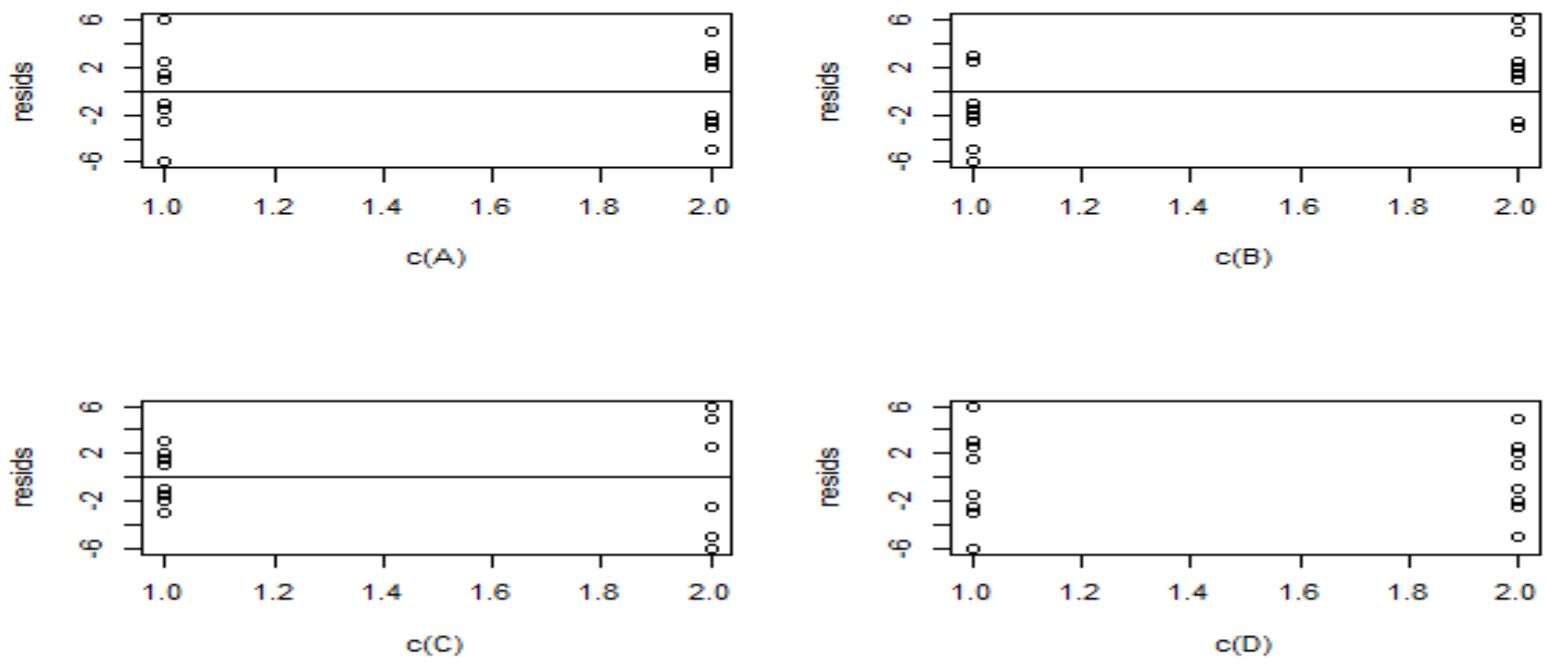
Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	1	1870.56	1870.56	83.3677	1.667e-05 ***
C	1	390.06	390.06	17.3844	0.0031244 **
D	1	855.56	855.56	38.1309	0.0002666 ***
A:C	1	1314.06	1314.06	58.5655	6.001e-05 ***
A:D	1	1105.56	1105.56	49.2730	0.0001105 ***
C:D	1	5.06	5.06	0.2256	0.6474830
A:C:D	1	10.56	10.56	0.4708	0.5120321
Residuals	8	179.50	22.44		


C
-1
1
mean of y
50
60
70
80
-1
1
A


D
1
-1
mean of y
60
70
80
90
-1
1
A


D
1
-1
mean of y
60
65
70
75
80
-1
1
C


resids
6
2
-2
-6
1.0
1.2
1.4
1.6
1.8
2.0
c(A)
c(B)
c(C)
c(D)

What Have We Learned?

We discussed $2^2, 2^3$ and 2^k factorial design in this chapter. We showed how the estimates of effects and ANOVA tables can be obtained by simple had calculations.

We also discussed three different widely used notations.